

What makes a protein degradable in *Klebsiella* and *E. coli*

Someone in the last meeting suggested that having a disordered region is a good sign a protein can be degraded. That is right — but only once you sharpen it, and only after you fix the bigger problem underneath it. Second half of this page: which papers and databases cover each part of the question, for **both** target organisms, and where the evidence runs out.

One thing up front: for this axis the usual direction is *reversed*. Every experimental dataset is *E. coli*, so *E. coli* is the organism with direct evidence and *Klebsiella* is the one receiving transfers. That is why *E. coli* leads on every measured row below — and why it is the one to build first.

WHICH PROTEASE? — SETTLED, AND IT CHANGES THE REST OF THIS DOCUMENT

The handle is **ClpP, activated directly, with no unfoldase partner** — named in the funded proposal, whose WP2 is called *WP2_ClpPELs*. Because activated ClpP **cannot unfold anything**, the initiation-region rule in §02 is an *unfoldase* rule and applies to the regime this project is *not* using. Under the Gr-ADI modality a substrate must already be unstructured — so **disorder is close to the whole requirement** rather than one aspect of six. Read §00 first.

IS DISORDER A GOOD INDICATOR?

For the **partnerless** handle this project uses — yes, and it is close to *the* criterion, because nothing pulls. What matters is **global** disorder plus a folded domain to put the ligand on (“two domains”). For the unfoldase channels the older, narrower answer still holds: disorder must be **terminal**, of a **specific length**, and **solvent-accessible**.

THE PROBLEM UNDERNEATH

Neither organism has ClpC or McsB — the absence is a property of Enterobacteriaceae. Every published BacPROTAC recruits ClpC, so the modality everyone cites does not transfer. The proposal's answer is to skip the unfoldase entirely and activate ClpP with chemistry.

WHAT DOES NOT WORK

Degron **motifs scanned on their own**. Two independent proteome-scale studies find no association between the classic recognition motifs and measured protein half-life. A motif only counts when the terminus carrying it is exposed.

THE DESIGN MISTAKE TO AVOID

Treating degradability as **one** property. Being *grabbable* and *staying gone* pull in opposite directions on natural half-life. Score them separately or the two cancel.

HOW SPARSE IS IT, REALLY?

Sharply split, and different per organism. **Reach, grab and recognise are solved at ~100% in both** — structural data already on disk, and the degron rules all got authoritative single-paper answers between 2024 and 2026. Everything *measured* favours *E. coli*, because Kp values must be transferred and only 55.5% of Kp proteins have a K-12 ortholog at all. The organisms diverge hardest on **assembly state** — experimental stoichiometry covers 1,779 *E. coli* proteins but only 30 in Kp — and on native proteolysis evidence, where Kp has 26 proteins from a single paper.

00 · THE ANSWER

Which protease: ClpP, activated directly, with no unfoldase partner

This document was first written while the recruiting end was treated as an open question. It is not one, and never was. The funded Gr-ADI research vision (Gr-ADI_ResearchVision_TPD-vs-Gram-Negs_Draft-v5(Final)) names the protease outright, and WP2 is called **WP2_ClPPELs** — ClpP-Engaging Ligands.

“...reduce target protein levels by **selectively engaging the bacterial ClpP protease**” · “Small molecule BacPROTACs that **activate the ClpP protease** and accelerate the rate of degradation of selected targets through a proximity effect”. ClpC is explicitly set aside: the cyclomarin-derived dCymM ligand “**only attaches to the Mtb ClpC**”, so the proposal pivots to the **ADEP** paradigm — direct chemical activation.

WHY “CLP” WAS CONFUSING, AND IT WAS NOT THE READER'S FAULT

The word names three different kinds of thing. **ClpP** is a peptidase that cannot unfold or select anything. **ClpX / ClpA / ClpC** are the ATP-driven unfoldase rings that give it specificity. **ClpB / ClpK** are disaggregases with no ClpP partner and are not degradation channels at all. **ClpQ/ClpY** are simply other names for HslV/HslU.

So *Klebsiella* and *E. coli* have exactly two Clp *channels* — ClpXP and ClpAP — and Gr-ADI recruits **neither**. It uses the shared peptidase alone, switched on chemically. Meanwhile every published BacPROTAC recruits **ClpC1**, which Enterobacteriaceae do not have, so the most visible prior art describes a machine these organisms lack.

The mechanism matters more than the name. ADEP-class activators bind the hydrophobic clefts where the ClpX/ClpA (L/I)GF loops normally dock, flipping ClpP from its compressed to its extended state: the axial pore opens and the catalytic triad orders. But **nothing pulls**. Activated ClpP has no motor — it cannot unfold, translocate, or select.

THE CONSEQUENCE: THE GATE IN §02 IS THE WRONG GATE FOR THIS PROJECT

The ~5 / ~20 / ~37-residue initiation-region thresholds this document is built on are an **unfoldase** rule: a loose end is enough *because the ring then does the unfolding work*. With no unfoldase, a substrate must **already be unstructured enough to diffuse into an open pore** — substantial intrinsic disorder, low conformational stability, a nascent chain, or a natively dynamic assembly cycle.

Read §02 and §03 as **Regime A** (with unfoldase: ClpXP, ClpAP, Lon, HslUV, FtsH) and this section as **Regime B** (partnerless activated ClpP — the Gr-ADI modality). Both are real; only one decides whether a target progresses.

Three things follow, and they promote what §04 already argued from a supporting observation to the central one:

1
Disorder is close to being the whole requirement under Regime B, not one aspect of six. The remark in the kick-off meeting that a disordered region might be a good indicator was correct — for this specific mechanistic reason.

2
“Partly disordered and partly ordered (two domains)” is the required architecture, not a preference: the disordered part is ClpP's entry point, the ordered part is where the target-engaging ligand binds with defined geometry. You need both, in the same protein. Computable as `two_domain_architecture` from the per-residue pLDDT profiles already held for 100% of both proteomes.

3
Conformational stability becomes a first-class feature — low melting temperature, marginal fold, high local unfolding propensity — and the repository currently has none.

INDEPENDENT CORROBORATION THAT THIS IS THE RIGHT MECHANISM

Jacques 2020 reports that the 200 most-depleted proteins under a ClpP activator are enriched for **ribosome-related** functions, and says the same held for ADEP4. Transferring both activator datasets onto our own proteomes puts **trigger factor** — the ribosome-associated chaperone — and a run of ribosomal proteins at the very top. That is the nascent-chain signature Regime B predicts, arrived at from a different direction.

Two bars, because the proposal's own criterion moved between drafts

SOURCE	WORDING
v3 draft	“show sensitivity to degradation by activated ClpP in the absence of an unfoldase partner ”

SOURCE	WORDING
v5 Final	“be a known, preferred substrate of a ClpP protease complex. ”
validation assay (both drafts)	“the activated ClpP of the target organism only (i.e. in the absence of an unfoldase partner) ” ... “ only targets that show degradation in this assay will progress ”
WP1 criterion (c) — what makes a good target — as worded in each draft, against the validation assay, which did not change.	

“A ClpP protease *complex*” admits ClpXP and ClpAP. So the **selection** bar is well populated — in *E. coli* essentially all ClpP substrate knowledge is ClpXP/ClpAP knowledge — while the **validation** bar that actually gates progression has almost no data. And they disagree systematically: a protein is a good ClpXP substrate *because* a motor grips and unfolds it, and many such proteins are stable, compact and folded. The score therefore emits `clpP_complex_substrate` and `partnerless_clpP` separately, plus a `bar_disagreement` flag, and **never averages them**. Targets that clear the written criterion and fail the assay proxy are the likeliest wet-lab disappointments, and naming them in advance is the point.

MEASURED: THE PROJECT'S FOUR NAMED TARGETS ON THE VALIDATION BAR

From the two activator datasets, transferred onto both proteomes. **AcpP 1.00 · DnaK 0.90 · GyrA 0.65 · GyrB 0.0015**. The first targets hold up. **GyrB has essentially no activated-ClpP evidence at all** — zero abundance change under ADEP4 ($p = 0.96$, dead-centre of the distribution), its single cleavage peptide negative and not significant, and absent from the ONC212 table entirely. It maps successfully at 44.8% identity, so this is absence of signal, not absence of mapping. GyrA/GyrB are the declared *second* targets.

Also worth flagging: v5 states both targets were “reduced by at least 2-fold upon treatment of either ClpP activator”. Under ONC212, DnaK is 1.4× and AcpP is **unchanged (+0.03)**. Both do have strong ONC212 *cleavage* evidence, so the conclusion survives by a better readout — but the stated justification does not.

WHAT THE CHOICE COSTS, AND ONE ADJACENT DELIVERABLE

The addressable target set is **much narrower** than the nine-channel survey below implies: a well-folded, stable, compact enzyme is a poor candidate however good its ligand. `clpP` is also **dispensable** in both organisms and ADEP resistance by `clpP` inactivation is documented, so the handle is itself a resistance liability. Human mitochondrial ClpP is a real off-target.

Separately, note that Ersilia's *other* named computational deliverable — the structure-based ClpP/activator pocket assessment in WP2_ClpPELs — concerns the **same structural object** as the mechanism above: the activator pocket *is* the ClpX/ClpA docking cleft. That work is out of scope for this document.

Six aspects, not three — and proteomics is not one of them

A natural way to carve this up is *disorder prediction*, *degron identification* and *proteomics experiments*. That gets a lot right, but it merges two things, misses three, and misfiles the third.

Disorder and degrons are two halves of **one** gate — is there a graspable start point, and does it carry a signal the machine reads. Proteomics is not a parallel category at all: it is the **ground truth** the other aspects get scored

against. And three aspects are missing entirely, two of which are the ones that actually kill targets.

GATE 1 · REACH

Can any protease here physically get to it?

Compartment decides the handle: cytoplasm → ClpXP, inner membrane → FtsH, periplasm and outer membrane → DegP. Get this wrong and nothing else matters.

missing from the 3-bucket model

GATE 2 · GRAB

Is there an accessible loose end, long enough to thread?

Per terminus, against 5 / 20 / 37 residues — plus internal disordered loops. Must be *terminal and exposed*, not globally disordered, and not buried in an oligomer.

= disorder prediction, sharpened

GATE 3 · RECOGNISE

Does that loose end carry a signal the machine reads?

Four separate rule sets — ClpXP C-degron, Lon C-degron, N-degron, FtsH membrane. Each has to be scored *through* the exposure of its own terminus.

= degron identification

GATE 4 · UNFOLD

Can the fold be pulled apart, and does it stay gone?

Unfolding is the rate-limiting step, so stability matters separately from disorder. And a protein the cell rebuilds fast is a poor target however grabbable it is.

missing — two aspects

Underneath all four: the validation layer. Measured turnover, half-lives and protease attribution are not a fifth aspect — they are the **labels**. They tell you whether a protein is *actually* degraded in vivo and by which protease, which is how you check the four gates are constructed correctly. This reframe has a practical consequence: it is why *E. coli* has to be built first, because that is where every label lives.

The gates are **serial and each can veto**. That is why a single blended score is misleading — a protein can look excellent on three gates and be untouchable on the fourth. Assembly state sits between gates 2 and 4 and is the sixth aspect: a sequence-disordered terminus facing the inside of a barrel is not a loose end.

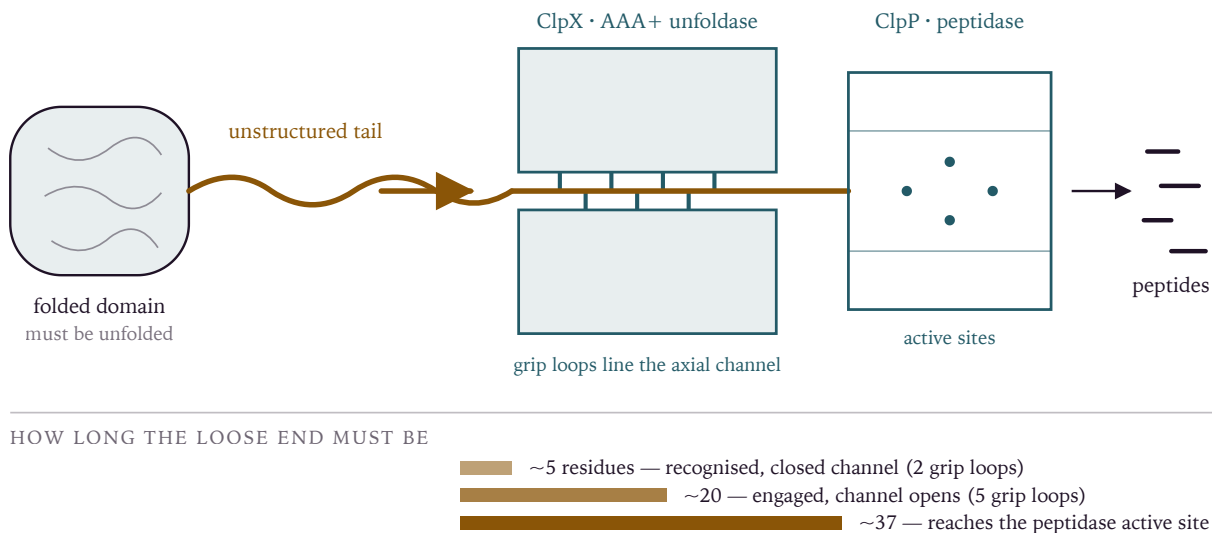
02 · MECHANISM

A protease has to thread the protein before it can destroy it

Bacteria have no ubiquitin. What they have instead are AAA+ proteases: a hexameric ATP-driven unfoldase ring stacked on a barrel-shaped peptidase. The ring grips a stretch of polypeptide with internal loops and pulls it

through a narrow axial channel hand over hand, unfolding whatever is attached and feeding it into the peptidase.

Everything about degradability follows from that one picture. The machine cannot grip a folded surface — it needs a **loose end** to start on. That loose end is called an *initiation region*, and its required length is a measured quantity, not a metaphor.



The reason a disordered terminus matters at all. Without a loose end there is nothing for the grip loops to hold, so the machine never starts. The three lengths are measured quantities from single-molecule and cryo-EM work on ClpXP, and they are why the feature has to be "how many accessible residues, at which terminus" rather than "is this protein disordered".

Two further consequences of the same picture, both easy to miss. First, **unfolding is the rate-limiting step** — so how thermodynamically stable the attached domain is matters as much as whether a tail exists. Second, a terminus buried inside an oligomeric assembly is not a loose end at all, however disordered its sequence looks.

03 · MACHINERY

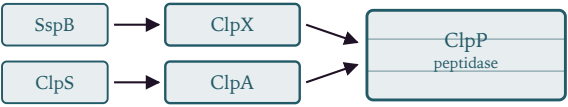
The machines, and how they differ

"Clp protease" is a family of combinations, not one enzyme, and the naming hides the architecture. **ClpP is only a peptidase** — a barrel with its active sites on the inside. It cannot unfold anything and cannot choose a substrate. What it does is get *rented* by different AAA+ unfoldase rings, and the ring is what does the choosing.

So **ClpXP and ClpAP are the same destruction chamber with a different selection engine bolted on.** That single fact explains most of the confusion in this area — and it is why *Klebsiella* and *E. coli* lacking ClpC costs them a selection engine, not a peptidase.

ATP-POWERED · PROGRESSIVE · NEEDS A LOOSE END

modular — one peptidase, several engines



adaptors



shared chamber — cannot unfold or select

ClpC — absent in Enterobacteriaceae

unfoldase rings — these choose

self-contained — unfoldase and peptidase in one unit, no adaptors



soluble

two chains

essential

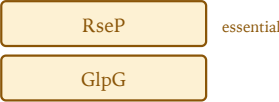
All of them **thread and destroy** — they grip a loose end, unfold the whole protein and translocate it in. This is where the 5 / 20 / 37-residue initiation region applies.

ATP-INDEPENDENT · ENDOPROTEOLYTIC

periplasm — serine proteases with PDZ domains



inside the membrane — cleave within the bilayer



These **cut, they do not thread**. No loose end is needed — what they need is an exposed cleavage site. So the initiation-region gate does not apply to this whole column.

Two architectures, not one. Everything on the left threads a loose end and destroys the whole chain; everything on the right makes cuts. That distinction matters more than any individual protease, because it means the “is there an accessible unstructured terminus” gate applies only to the left-hand column — a simplification worth correcting if you have been thinking of it as universal.

Similarities and differences, machine by machine

MACHINE	ARCHITECTURE	WHERE	HOW IT PICKS SUBSTRATES	COMPUTABLE RULE	ESSENTIAL · PRECEDENT
ClpXP	ClpX unfoldase ring + shared ClpP barrel	cytoplasm	ClpX reads terminal degrons directly; SspB adaptor delivers ssrA-tagged substrates	DEtox C-degron matrix [L/F/Y/W]–[L/A/R]–L–A–A; Flynn N-motifs	not essential · two handles : ClpX zinc-binding domain (CLIPPERS) and SspB-adaptor hijack (bacNID, 2025)
ClpAP	ClpA unfoldase ring + the same ClpP barrel	cytoplasm	ClpS adaptor switches ClpA onto N-degron substrates — and simultaneously <i>blocks</i> ssrA delivery	N-degron P1–P5 model; ClpS pocket fits Leu/Phe/Trp/Tyr	not essential · prior only
Lon	unfoldase + peptidase in one chain ; no adaptors	cytoplasm	reads its own degrons: buried aromatic clusters exposed on misfolding	x–[L/I]–[L/I/V]–H–C00H, conserved across enterics	not essential · engineered handle (orthogonal light-gated mLon in E. coli, 2026)
HslUV	HslU unfoldase + HslV peptidase — a separate pair, closest thing here to a proteasome	cytoplasm	poorly characterised; exposed aromatic or cationic residue in a disordered segment	none usable — degenerate with the Lon feature	not essential · <i>dropped from the axis</i>

MACHINE	ARCHITECTURE	WHERE	HOW IT PICKS SUBSTRATES	COMPUTABLE RULE	ESSENTIAL · PRECEDENT
FtsH	self-contained, and anchored in the membrane	inner membrane	pulls substrates out of the bilayer; needs a cytoplasmic tail, or senses misfolding directly	tail >20 aa (N) / ~10 aa (C); lipid-facing polar residues inside TM helices	essential (expt.) · prior only
DegS	ATP-independent serine protease, trimeric, PDZ-regulated	periplasm	allosterically switched on by a peptide binding its PDZ domain — a natural molecular glue	C-terminal ϕ -x-Phe, exposed only when an OMP misfolds	essential (expt.) · prior only
DegP	ATP-independent; switches between chaperone and protease	periplasm	binds misfolded substrate; activity tunable by small molecules	cleaves between paired hydrophobic residues ; substrate-enriched dipeptides	not essential · demonstrated (glues, preprint)
Prc / Tsp	ATP-independent, C-terminal-specific	periplasm	reads free C-termini — including ssrA tags on proteins that got exported	non-polar C-terminal tail	not essential · prior only
RseP	intramembrane <i>metallo</i> protease (site-2 family)	in the bilayer	unwinds a TM helix by strand addition and clamps it at the active site	helix-destabilising residues within the TM segment ; exposed C-terminal hydrophobe	essential (expt.) · prior only
GlpG	intramembrane <i>serine</i> protease (rhomboid)	in the bilayer	a sequence motif at the cleavage site matters <i>more</i> than helix destabilisation	small side chain at P1, negative charge at P1' — authors state genome-wide prediction is feasible	not essential · prior only
Essentiality is from our own <i>E. coli</i> composite (07h), experimental where marked. “Precedent” asks whether anyone has actually built a degrader that recruits this machine — only two have been, which is why the rest are scored as mechanistic priors.					

Klebsiella versus *E. coli*: nearly the same machine shop

This matters because it decides how much transfers. The answer is: almost everything. Both are Enterobacteriaceae, both lack ClpC and MscB, and every one of the nine channels is present in both — at near-identical length, which is the real signal.

MACHINE	E. COLI K-12	KP HS11286	READ
ClpX	P0A6H1 · 424	A0A0H3GSR4 · 424	Identical length. The handle CLIPPERS binds is the same protein in both.
FtsH	P0AAI3 · 644	A0A0H3GTR0 · 644	Identical. Essential in both (expt.).
RseP	P0AEH1 · 450	A0A0H3GMW3 · 450	Identical. Essential in both (expt.) — Kp 0.82, Ec 0.88.
GlpG	P09391 · 276	A0A0H3GUB5 · 276	Identical.
Prc / Tsp	P23865 · 682	A0A0H3GQ33 · 682	Identical.

MACHINE	E. COLI K-12	KP HS11286	READ
DegS	P0AEE3 · 355	A0A0H3GY66 · 352	Essential in both (expt.) — Kp 0.60, Ec 0.88.
ClpA · ClpS	P0ABH9 · 758 P0A8Q6 · 106	A0A0H3GQX4 · 759 A0A0H3GKY1 · 105	Both present; ClpS dispensable in both, so the N-degron route is switchable off in either.
ClpP	P0A6G7 · 207	A0A0H3GKH6 · 194	Kp entry is 13 residues shorter — almost certainly an N-terminal annotation truncation, not biology. Worth checking before any structural work.
Lon	P0A9M0 · 784	A0A0H3GJ60 · 741	43 residues shorter in Kp; same caveat.
ClpK	— absent —	A0A0H3GSF4 · 884	The one real machinery difference. A Kp-specific Hsp100 giving thermotolerance and nosocomial persistence. But it is ClpB-class with <i>no ClpP partner</i> , so it is not a degradation channel — a disaggregase, not a protease.
OmpT	P09169 · 317	— absent —	An outer-membrane protease Kp lacks. Not a handle, but a genuine difference in the envelope.

Lengths in residues, measured from the two reference proteomes on disk (UP000000625, UP000007841). Essentiality from our own composites; “expt.” means an experimental screen, not a prediction. **Every pairing here was verified by pairwise sequence alignment** — not taken from an ortholog table — at 76–99% identity, against a 20–32% random-pair baseline.

WHAT ACTUALLY DIFFERS, AND WHAT IT COSTS

The biology barely differs. Nine channels in both, five of them at byte-identical length, and the same three essential machines — **FtsH, DegS and RseP**. Every Clp, Lon and HslUV component is dispensable in both, so the resistance argument is identical too. This is why rules derived in *E. coli* can be carried to *Klebsiella* with a clear conscience.

What differs is the annotation, and it bites. *E. coli* K-12 is a curated Swiss-Prot reference proteome where everything has a gene name. Kp HS11286 is TrEMBL: **eleven of these machinery genes carry no gene symbol at all** — ClpA, SspB, DegP, DegQ, DegS, RseP, Prc, SohB, PtrA, PtrB and ClpK — and had to be identified by orthology or locus tag. ClpK exists in Kp only as KPHS_23030. Any pipeline step that greps gene symbols will silently miss most of the Kp machinery, which is a mistake already made once in this project.

And the evidence differs enormously, which is the real reason to start with *E. coli* — not the machinery, but everything measured about its substrates.

ClpXP gained a second handle. Alongside CLIPPERS binding the ClpX zinc-binding domain, a 2025 *Nat Commun* paper hijacks the **SspB adaptor** instead, displaying both a target-binding and an SspB-binding peptide on a nanoparticle. Two distinct entry points to the same machine.

Lon gained its first. An orthogonal, light-gated *Mesoplasma florum* Lon now works in *E. coli* and — usefully for us — degrades in **stationary phase**, which decouples measured degradation from growth dilution.

FtsH degradability is gated, not intrinsic. FtsH sits inside an **HflK/HflC cage** whose aperture is condition-dependent; cells locked shut fail to recover from aminoglycosides. So a membrane protein's FtsH score needs an accessibility term, not just a degron.

For ClpAP, unfolding is rate-limiting — not translocation. Measured directly: ClpAP unfolds roughly $3\times$ faster than ClpA alone, and translocation is an order of magnitude faster than unfolding either way. So the ClpAP term should be dominated by local mechanical stability next to the degron, not by chain length.

DegP is redox-gated, and Prc may act first. DegP's protease activity depends on the redox state of a disulfide in its own protease domain and on whether the substrate carries free cysteines. Separately, in-cell NMR shows **Prc cleaving a metallo- β -lactamase first, with DegP then processing the fragments** — so for β -lactamases the periplasmic route may be Prc→DegP rather than DegP alone.

THE THREE THINGS WORTH TAKING AWAY

One. ClpP is a rented chamber. ClpXP and ClpAP differ only in the engine and the adaptor, so a handle that grabs ClpX does not automatically reach ClpA substrates, and vice versa. ClpS is not merely an adaptor — it *reassigns* ClpA and blocks the ssrA route while it does so.

Two. The loose-end requirement is a property of the *left-hand* architecture only. For DegS, DegP, Prc, RseP and GlpG the question is not “is there a threadable terminus” but “is there an exposed cleavage site”. Scoring the periplasm on initiation regions would be a category error.

Three. The three machines that *cannot* be deleted to escape a degrader — FtsH, DegS and RseP — are all in the envelope, and none of them has any degrader precedent. The two machines with precedent, ClpXP and DegP, are both dispensable. That tension is the most interesting thing in this table, and it is an argument for looking hard at DegS: essential, and already known to be switched on by a C-terminal peptide.

Disorder is the right instinct in four specific ways

1

It has to be at a terminus

Won et al. (2024) screened 72 native *M. smegmatis* proteins for chemically induced degradation and fitted a model over 485 sequence and structure descriptors. The single top-ranked feature was the **mean disorder propensity of the N-terminal 30 residues**. The C-terminal 30 residues showed no correlation. Their phrasing: "*the more disordered the N-terminal sequence is, the more efficiently TPD could act.*"

Two caveats worth carrying. That study used ClpC1, which reads N-terminal signals; ClpX reads *both* termini, so we should measure each terminus separately rather than import the asymmetry. And internal disordered loops also work — *E. coli* FtsZ has a genuine ClpXP site inside its linker at residues 349–358 (QE_QKPVAKVV).

2

It holds up at proteome scale

Gupta et al. (2024) measured turnover for 3,262 *E. coli* proteins across 14 conditions. Disordered proteins had significantly shorter half-lives, at $p \approx 1 \times 10^{-208}$. Also short-lived: proteins under 10 kDa, transcriptional regulators, and iron–sulfur-cluster proteins.

The same dataset carries a compartment result that reshapes everything downstream: cytoplasmic proteins are selectively degraded, while **membrane and periplasmic proteins are largely stable**.

3

Length and position are measurable, so measure them

The thresholds in the figure above give a real scale: about 5 residues to be recognised, 20 to be properly engaged, 37 to reach the peptidase — and beyond that, longer tails keep raising the odds that a grip attempt commits instead of slipping. Degradation rate even depends on *which* terminus is pulled, because the unfolding force required differs by geometry.

Usefully, we can compute all of this today at no cost: AlphaFold models are already on disk for 5,727 of the 5,728 *Klebsiella* proteins and 4,371 of the 4,403 *E. coli* ones, and per-residue pLDDT plus solvent accessibility over the terminal windows gives exactly these features.

4

Accessible, not merely unstructured

A disordered sequence at the mouth of a barrel interior is useless. So the feature is *disorder* × *solvent exposure* × *not buried in an assembly*, and only then does a recognition motif on that terminus count for anything.

Disorder is not a universal accelerant of proteolysis — disordered regions can be context-dependently protease-*resistant*. The argument above is specific to ATP-dependent processive AAA+ proteases, where an accessible initiation region is a mechanical requirement.

More pointedly: **motifs scanned on annotated protein termini do not predict half-life**. Nagar et al. (2021) fitted half-lives for 1,149 *E. coli* proteins over 188 features and found the most informative were protein-interaction-network properties — connectivity, not sequence — with no significant association for the N-end rule or the classic ClpXP motifs. Gupta 2024 independently reports no enrichment of destabilising N-terminal residues among fast-degrading proteins.

The likely reason is that real degrons are usually **latent**. Humbard et al. (2013) pulled over 100 degron-bearing *E. coli* proteins off a native lysate and found most had been trimmed by other proteases first — the degron did not exist in the intact protein. LexA is the textbook case: its self-cleavage fragment's *new* C-terminus is a bona fide degron that is absent from the full-length protein. Scanning the annotated start and stop of an open reading frame therefore misses much of the real signal.

WHAT THE NUMBERS IN THE APP CURRENTLY DO

Worth knowing before anyone reads the existing degradability column. The current score is a set of sequence patterns, and its dominant feature is **implemented backwards**.

It flags a protein as degradable when residue 2 is bulky — leucine, phenylalanine, tyrosine or tryptophan. But the N-end rule needs a bulky residue at position **1 of the mature protein**, and methionine aminopeptidase removes the starting methionine *only when residue 2 is small*. A bulky residue 2 therefore means the methionine **stays**, and the mature N-terminus is methionine — which is stabilising. Checked across all 5,728 proteins: every one of the 684 flagged has a bulky residue 2, so the feature selects precisely the proteins that do *not* expose a degron. And it supplies **680** of the 698 mid-tier calls.

Two smaller faults in the same set: the pattern meant to detect ssrA-like tails **cannot match the ssrA tag itself** (the tag ends ...YALAA, which the alternating pattern rejects), so it fires on 5 proteins out of 5,728; and one N-terminal pattern matches 30.7% of the proteome, which cannot discriminate anything. **Treat the current degradability column as a placeholder, not a measurement** — and note that the *E. coli* values in the app are an explicit mock, flagged as provisional.

Klebsiella does not have the machine the field publishes about

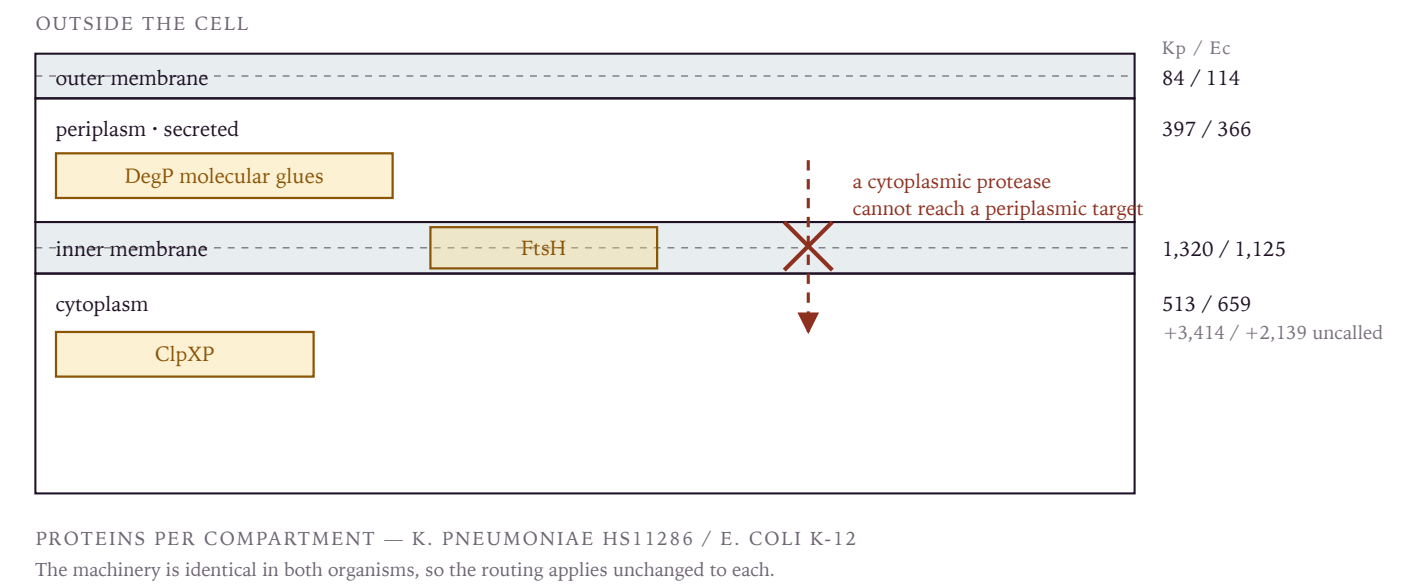
BacPROTACs — the entire published small-molecule bacterial-degrader literature — recruit **ClpC1** using a cyclamarin-A-derived ligand, and the natural tag that system reads is phosphoarginine, written by the kinase **McsB**. Both are Firmicute and Actinobacteria proteins.

We checked this properly, by enumerating the whole Hsp100 (ClpA/ClpB) family in each proteome rather than trusting gene names, which are patchy in the *Klebsiella* HS11286 annotation:

PROTEOME	HSP100 MEMBERS FOUND	CLPC?
<i>K. pneumoniae</i> HS11286	ClpA (759 aa), ClpB (823), ClpK (884)	none
<i>E. coli</i> K-12	ClpA (758), ClpB (857)	none

PROTEOME	HSP100 MEMBERS FOUND	CLPC?
<i>B. subtilis</i>	ClpC (810), ClpE (699)	yes
<i>M. tuberculosis</i>	ClpC1 (848)	yes
Hsp100 family census by proteome. <i>B. subtilis</i> and <i>M. tuberculosis</i> are positive controls. Also absent from both Enterobacteriaceae: clpE, clpL, mcsA/mcsB, the pupylation genes (pafA/pup/mpa/dop), and any 20S proteasome subunit.		

So the chemotype has no receptor in *either* of our target organisms — the absence is a property of Enterobacteriaceae, not of *Klebsiella* specifically, so everything that follows applies equally to both. What they do have is the *E. coli*-like set — ClpXP, ClpAP with its ClpS adaptor, Lon, HslUV, FtsH, the SspB adaptor, and the periplasmic DegP family — and each of those acts in a particular compartment. That turns degradability from one question into three.



Which protease can reach a protein is decided by where the protein lives. This is a categorical routing decision, not a scoring penalty — and the crossed arrow is a real published mistake, not a hypothetical one (see below).

The cytoplasmic handle is real, and recent

Izert-Nowakowska et al. (2025) built **CLIPPERS**: a bait–linker–anchor peptide whose anchor is the **XB motif of SspB**, which binds the zinc-binding domain of ClpX. They degraded endogenous, untagged, degron-free GroEL in living *E. coli*, and the effect vanished in ΔclpX and ΔclpP. This is the first demonstration of targeted degradation of a native protein in a Gram-negative, and it is what makes this axis defensible at all. Be honest about magnitude, though: about **17%** depletion at one hour and **40%** at six, with 20–25% growth inhibition — and it is a plasmid-expressed peptide, not a drug.

The periplasmic handle is the most drug-like result in the field

Taylor et al. (2026) found that **tazobactam itself accelerates DegP-mediated degradation of TEM β-lactamases**, and improved it further by grafting on dipeptide motifs enriched among DegP substrates — with no

linker at all. A molecular glue rather than a chimera: small, orally dosable, and acting in the compartment the target actually occupies. It means the periplasm and outer membrane are not a write-off; they are a separate, high-value sub-axis, containing among other things the β -lactamases.

WHY COMPARTMENT IS A HARD FILTER, NOT A WEIGHT

One group reported a small-molecule "BacPROTAC" against CTX-M-14 β -lactamase, made by conjugating nacubactam to the full *ssrA* degron. It has two problems. It weighs **1,669** daltons, which is a lot to push through a Gram-negative outer membrane. And CTX-M β -lactamases are **periplasmic while ClpXP is cytoplasmic** — so to make the experiment work, the target was engineered to stay in the cytoplasm.

A compartment mismatch is not a penalty to be traded off against a good score. It is a category error, and the scoring has to make it impossible by construction.

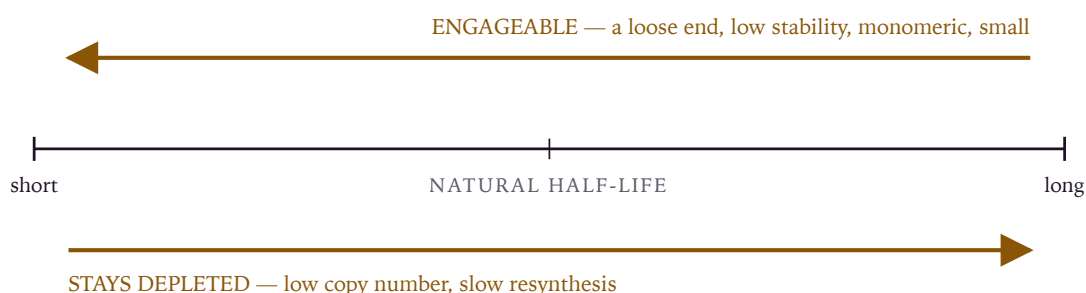
One thing to say out loud about resistance

In *Klebsiella*, *clpP*, *clpX*, *lon*, *hslUV* and *clpS* are all dispensable — $\Delta clpX$ and $\Delta clpP$ mutants grow like wild type. That cuts both ways. It is what we want in a handle, because then the lethality comes from destroying the target rather than from poisoning proteostasis. But it also means the bacterium can escape simply by losing the protease we recruit. *ftsH* is the one exception — genuinely essential — but it is membrane-anchored, so its reach is limited.

06 · UNFOLD & STAY DOWN

"Degradable" is two questions, and they disagree

This is the part that is easiest to get wrong in a score, because both halves sound like the same thing.



The same measurement is evidence for one requirement and against the other. A single "fast turnover means good target" term therefore cancels itself. The fix is to source engageability from structure, termini and stability, and to carry resynthesis burden as a separate column used for filtering rather than folded into the score.

The evidence for the right-hand arrow is quantitative, not intuitive. Won 2024 found that higher steady-state abundance *anti*-correlates with induced degradation rate at $r = -0.69$, $P = 6.6 \times 10^{-9}$; the CLIPPERS authors say the same thing independently — high-copy-number targets resist. Degradation is a flux competition against resynthesis, so a protein the cell rebuilds quickly is a poor target no matter how grabbable it is.

GroEL fails on both counts, and it fails for reasons you can compute. Both its N- and C-termini **face the interior of the chaperonin barrel**, so there is no accessible loose end unless a subunit dissociates. And it is one of the most abundant proteins in the cell, so anything degraded is replaced fast.

It is exactly the protein CLIPPERS could only deplete by about 40%, and the authors concluded that what they degraded was mostly newly translated, not-yet-assembled subunits. **Any implementation of this axis that ranks GroEL highly is wrong** — which makes it a genuine test rather than a story.

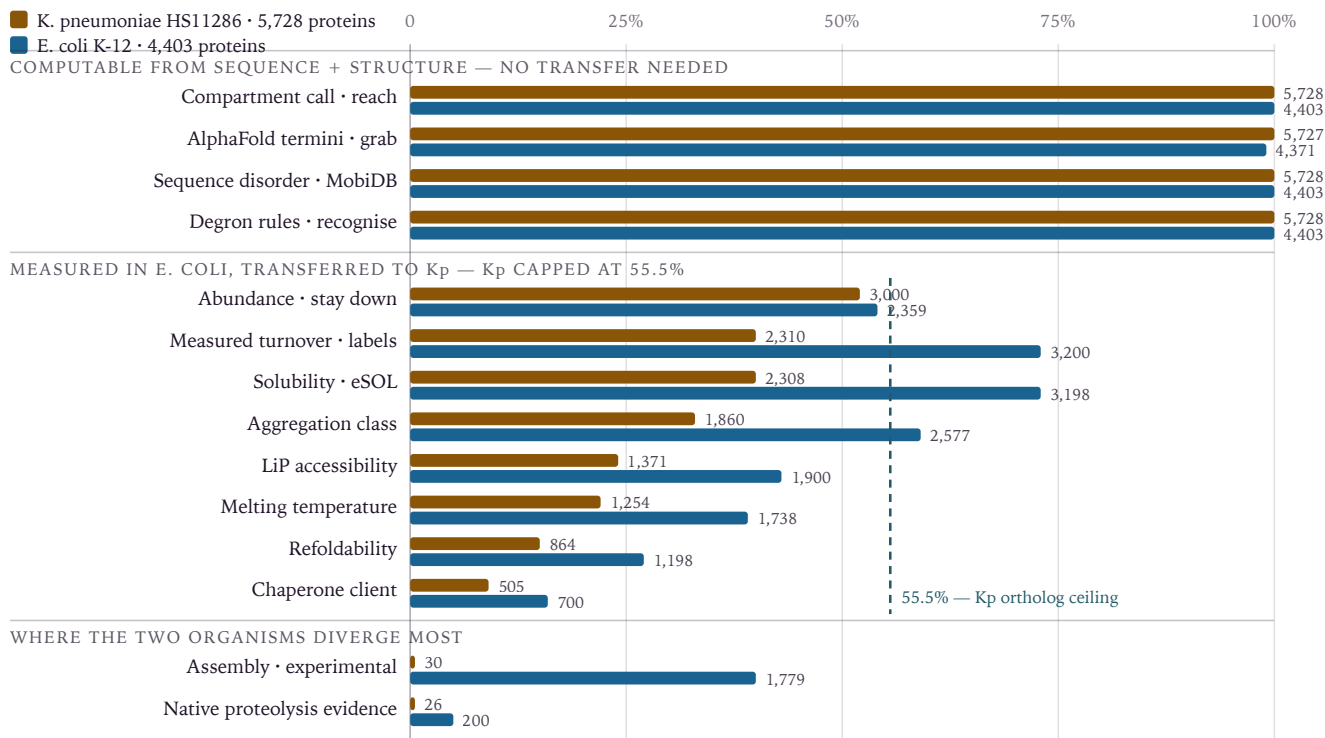
One more practical consequence of degradation being *partial*: prefer targets with a steep dose-response. A protein that only matters when fully absent is a bad fit for a modality that removes 40% of it. We already have that information — CRISPRi knockdown vulnerability from the essentiality axis measures precisely this.

07 · COVERAGE

Where the evidence actually is

This is the part that decides how much work each aspect costs. For every aspect, how many proteins would end up with a real value — out of 5,728 in *K. pneumoniae* HS11286 and 4,403 in *E. coli* K-12?

The asymmetry is structural, not incidental. *E. coli* receives every measured dataset directly. Kp receives it only through orthology, and only **3,179** (55.5%) of Kp proteins have a K-12 ortholog — a hard ceiling no dataset can beat. So *E. coli* leads on every measured row, and the gap is simply the cost of working in the pathogen rather than the model.



Proteins that would carry a real value. E. coli figures are each dataset's own coverage; Kp figures are estimates — dataset size × ortholog rate. The ceiling binds only the transferred band: E. coli needs no transfer, which is why it leads on every measured row. "Assembly · experimental" is PDB-derived stoichiometry only — Seq2Symm predicts symmetry from sequence for 100% of both proteomes.

The split is sharp, and it is not where you would guess. The three gates people worry about most — is it disordered, does it have a degron, can a protease reach it — are already at full coverage in *both* organisms. What is thin is everything about *unfolding*. And the two bottom rows are the ones to notice: assembly state is routine in *E. coli* and effectively hopeless in *Klebsiella*, the single largest asymmetry between them.

Aspect by aspect — organism, source, and why it matters

ASPECT	KP / EC COVERAGE	SOURCE	WHY IT MATTERS
1 • Reach compartment → handle SOLVED	Kp 5,728 100% Ec 4,403 100%	DeepLocPro (Bioinformatics 2024) + PSORTdb + STEPdb 2.0 — already built in-repo as 09a–09h	Decides which protease can physically touch the protein: cytoplasm → ClpXP, inner membrane → FtsH, periplasm and OM → DegP. A compartment mismatch is a <i>category error</i> , not a scoring penalty — it is what invalidated the one published Gram-negative degrader.
2 • Grab the loose end SOLVED, FREE	Kp 5,727 100% Ec 4,371 99%	AlphaFold pLDDT + solvent accessibility on the models already on disk ; window recipe from Won 2024 ; MobiDB bulk for sequence disorder. And for the <i>real</i> terminus rather than the annotated one: TRAINSPOTTER (1,082 measured Ec N-termini) and the PeptideAtlas coordinate map	An unfoldase cannot grip a folded surface — it needs an accessible unstructured terminus long enough to thread (5 / 20 / 37 residues). N-terminal-30 disorder is the strongest published predictor of bacterial degradation efficiency.

ASPECT	KP / EC COVERAGE	SOURCE	WHY IT MATTERS
3 • Recognise degron rules GOOD (2024–26)	Kp 5,728 100% Ec 4,403 100%	Nine rule sets , one paper each. <i>Cytoplasm</i> : DEtox (ClpXP C-degron matrix) • Cragan 2025 (Lon C-degron) • Sen 2025 “N-FIVE” (N-degron P1–P5). <i>Membrane</i> : Chai-Danino 2026 (FtsH lipid-facing TM polars) • RseP (helix-destabilising TM residues) • GlpG (small P1 / negative P1'). <i>Periplasm</i> : DegS (C-terminal ϕ -x-Phe) • DegP (paired hydrophobics) • Prc (free C-terminus)	A loose end still needs a signal the machine reads. But a motif only counts <i>multiplied by</i> the exposure of its own terminus — scanned alone, the classic motifs show no association with measured half-life.
Validation the labels GOOD FOR E. COLI	Kp ~2,310 40% Ec 3,200 73%	Gupta 2024 Table S1 — 3,262 proteins $\times$ 14 conditions , UniProt-keyed, and those conditions include $\Delta clpP$, Δlon , $\Delta hslV$, the triple KO and $\Delta smpB$. Then Nagar 2021 (feature matrix), MacKrell 2026 (stationary phase)	The only ground truth. And note what those knockout columns really are: a per-protein change in half-life when you delete a specific protease is a label for that channel, not a feature . This is the single reason <i>E. coli</i> has to be built first — it is the only organism in which this axis can be trained or validated at all.
4a • Unfold can it be pulled apart THIN: POOL 6–7	Kp 505–2,308 9–40% Ec 700–3,198 16–73%	No single source works . Pool Cappelletti LiP • Mateus 2018 Tm • Mateus 2020 stability variance • eSOL • Györkei aggregation • To 2021 refoldability • Calloni chaperone clients • Onyenemezu 2026 acid-stability pH50 (1,675 proteins, and it stratifies by compartment). PHDB may bundle several	Unfolding is the <i>rate-limiting step</i> of AAA+ degradation, so a stable domain resists even with a perfect handle. There is no proteome-scale folding-stability measurement for any bacterium, which is why this row has no single answer.
4b • Stay down resynthesis flux SOLVED	Kp ~3,000 52% Ec 2,359 54%	Li 2014 absolute <i>synthesis rates</i> — molecules per cell per generation, 3,041 high-confidence — which is the real denominator, not abundance • Schmidt 2016 copies/cell • Illenseher 2025 Kp-native iBAQ	Degradation is a flux competition against resynthesis: higher abundance measurably slows induced degradation ($r = -0.69$). Kept as its own column rather than folded into the score, because it pulls <i>opposite</i> to “grab” on half-life.
Assembly state is the terminus free SOLVED IN EC WEAKEST IN KP	Kp 30 0.5% Ec 1,779 40%	RCSB assembly API — oligomeric_state returns literally “Homo 2-mer” across 7,671 Ec assemblies • PDBe interface residues , in UniProt numbering, from <i>biological assemblies</i> , with the partner named • Complex Portal (324 complexes, explicit stoichiometry) • Seq2Symm for 100% coverage from sequence alone • STRING physical subnetwork for connectivity. <i>Not</i> the Schweke atlas — it is O157:H7, with zero accession overlap with K-12	A terminus facing the inside of an oligomer is not a loose end however disordered its sequence looks — exactly why GroEL resisted. An earlier version of this page had it wrong : this is not the weakest aspect, it is the weakest aspect <i>in Kp</i> . For <i>E. coli</i> , PDBe returns terminal burial in the assembly <i>with the partner named</i> , so self-burial in a homomer is separable from burial against a different subunit — the GroEL feature, computed directly. Still the largest Kp/Ec asymmetry here.
Native evidence measured in the organism ABSENT IN KP	Kp 26 0.5% Ec ~200 5%	Kp: Muselius 2020 Δlon . Ec: Gupta protease attribution plus the Flynn, Neher and FtsH trap sets and the N-degradome	The only organism-specific check that the transfers are not systematically wrong. Kp has essentially none — no meltome, no LiP-MS, no turnover, no ClpP structure — the honest limit of the whole axis.

Coverage is proteins that would carry a real value, of 5,728 (Kp HS11286) and 4,403 (Ec K-12). Kp figures on transferred rows are estimates — dataset size × ortholog rate. Full sourcing with exact filenames: docs/degradability_downloads.md.

WHY UNFOLDABILITY CANNOT BE FIXED BY FINDING A BETTER PAPER

There is **no proteome-scale experimental folding-stability dataset for any bacterium**. The obvious candidate — the 2023 mega-scale study with ~776,000 stability measurements — covers 331 natural domains of 40–72 residues, selected for a display assay and not mapped to any proteome. Using it would mean training a predictor, which converts an experimental column into a predicted one.

So melting temperature is the practical stand-in, and the honest approach is to pool several mid-coverage measurements that each capture a different facet — thermal stability, refoldability, solubility, aggregation, chaperone dependence, protease accessibility — and accept that any given protein will have two or three of them rather than all seven. This is the one aspect where “*maybe I need a lot more because it is sparse*” is exactly right.

08 · SHORTLIST

If you only fetch six things

Revised for an *E. coli*-first build. These six take the axis from unbuilt to trainable: one label set, the real protein termini, the three cytoplasmic rule sets, and the single most predictive feature class. All open access, all scriptable.

#	SOURCE	WHERE TO GET IT	WHAT IT GIVES
1	Gupta 2024 <i>Nat Commun</i> 15:5890	PMC11246515 → MOESM4, MOESM5, MOESM7	3,262 proteins × 14 conditions with degradation rates and protease attribution, plus ~600 measured N-termini. The label set everything else is checked against.
2	DEtox Beardslee & Schmitz 2024	PMC10862746 → media-2.xlsx, media-3.xlsx	The measured ClpXP C-degron matrix, from ~100k tags. Replaces a regex that cannot match the tag it was written for.
3	Cragan 2025 <i>JBC</i> 301:108365	PMC11986505 → mmc1.xlsx	The Lon C-degron rule — a channel we currently have nothing for, and one that transfers to Kp because it is conserved across enterics.
4	Sen 2025 “N-FIVE” preprint — flag it as such	github.com/KunjapurLab/N-terminal-cluster-stability	The N-degron P1–P5 model. Note the data and model are in the repo, <i>not</i> in the paper's supplement.
5	TRAINSPOTTER Van Damme 2026, NAR	PMC13245402 → Europe PMC zip → Table S1.xlsx	1,082 measured <i>E. coli</i> N-termini across 729 proteins, with initiator-Met excised vs retained. The N-degron channel is otherwise scored on annotated start codons that are frequently wrong.
6	STRING v12 taxon 511145, not 83333	stringdb-downloads.org/.../511145.protein.physical.links.v12.0.txt.gz	Per-protein interaction degree at 91% coverage . Nagar 2021 found network connectivity outranked every sequence motif for predicting half-life — so this is the highest-value feature we do not already have.

Ordered by what unblocks the most: the label set, then the real termini, then the rules, then the strongest feature. eSOL and the MobiDB proteome bulk download — both a single URL, both trivial — are next after these; full

THE ONE URL PATTERN WORTH KNOWING

For any open-access paper, this returns a zip of every supplementary file:

ebi.ac.uk/europepmc/webservices/rest/<PMCID>/supplementaryFiles

Verified working for 15 of the papers behind this page. It avoids two traps: publisher sites reject scripted requests (pnas.org, cell.com, jbc.org, Wiley and ACS all did), and the direct PMC file URLs return a placeholder page instead of the file.

09 · WHAT THIS MEANS

What this means for the target list

Put together, the axis becomes a routing decision followed by four weighted channels, rather than a bag of sequence regexes. In plain terms, what we now ask of each protein:

- Which compartment is it in, and therefore **which protease could ever reach it**? Cytoplasm sends it to ClpXP, inner membrane to FtsH, periplasm and outer membrane to DegP.
- Does it have an **accessible unstructured terminal region** of usable length — measured per terminus from the AlphaFold model, against the 5 / 20 / 37-residue scale?
- Does that accessible terminus carry a **recognition motif** the relevant machine actually reads, with the adaptor it needs present in the genome?
- Is its *E. coli* counterpart **measurably unstable in vivo**, and if so, is a known protease responsible?
- Is it **physically unfoldable** — low melting temperature, poorly refoldable, aggregation-prone, chaperone-dependent, not knotted, not locked in an assembly?
- Separately: what is the **resynthesis burden**? Reported as its own number, used to filter, never mixed into the score.

AND THE REASON TO CARE IN THE FIRST PLACE

Degradation is event-driven rather than occupancy-driven, removes scaffolding and non-catalytic function, and needs no active-site pocket. So its *unique* value sits on proteins that are **essential but poorly ligandable** — the targets where a conventional inhibitor has nowhere to bind. That is a quadrant we already score on two other axes, so the prioritisation now emits it directly as a column rather than leaving it to be inferred from two sliders.

Honest limits: no *Klebsiella* protein has ever been chemically degraded, and there is no *Klebsiella* turnover, melting-temperature or limited-proteolysis dataset — every biophysical column has to be transferred from *E. coli*, which caps coverage at 3,179 of 5,728 proteins. Delivery of a degrader into a Gram-negative cytoplasm remains unsolved. This axis is a mechanistic prior, and should be presented as one.

Detail layer, in the repository: docs/03_degradability.md (the specification) · docs/degradability_report.md (audit, dataset inventory, validation plan) · docs/degradability_downloads.md (the download checklist, with exact filenames and verified routes) · docs/degradability_references.md (every citation).

Open decisions, and what is honestly missing

Two versions of the score now exist

A second agent has been implementing this axis in parallel, and has already landed the whole localisation layer (09a–09h) plus the beginnings of the degradability pipeline (`src/degradability.py`, 10a, 10b). Its audit independently found the three bugs described above; I re-derived each one against the data before believing it, and they hold.

The two designs agree on the things that matter — structural exposure weighted above sequence motifs, a motif contributing only through the exposure of its own terminus, and missing measurements renormalised away rather than scored as zero. They differ on two real points, and someone should choose rather than letting both drift:

- How the composite is organised.** Their version weights by *evidence provenance* — own-organism measurement 0.45, ortholog transfer 0.35, sequence prior 0.20. The specification weights by *mechanism* — engagement, degron, turnover, biophysics. Provenance weighting is more honest about confidence; mechanism weighting is more diagnosable when a target scores badly. They are not reconcilable by averaging.
- What happens to non-cytoplasmic proteins.** Their version applies a minimum Clp-accessibility threshold, which *excludes* them. The specification treats compartment as a *router* and sends them to FtsH or DegP instead. This is the bigger decision: it is the difference between scoring 4,200 proteins and scoring all 5,728, and it decides whether the β -lactamases are in scope.

One thing from their side that should win outright: they set motif weights from the **measured enrichment of each motif against real turnover data**, with a motif whose odds ratio sits near 1 weighted to zero. That is strictly better than the hand-assigned weights in the specification, and it is exactly the discipline whose absence produced the inverted feature in the first place.

Two things to design around

A COMPETITOR EXISTS, AND ONE PUBLISHED RESULT MUST BE EXCLUDED

ProHL (*J Microbiol Biotechnol* 2026) is a deep-learning classifier of bacterial protein half-life, validated in *E. coli*. It is the closest thing to a competitor for the predictive part of this axis. Read its *corrigendum* before benchmarking against it. Our differentiator should not be half-life prediction — half-life is an *input*. It is per-channel, compartment-routed *recruitability*, with the demonstrated-versus-prior distinction kept explicit.

LAMP-D (*JACS* 2025) degrades NDM-1 in a Gram-negative with 53-fold meropenem potentiation, and would look like an ideal positive example. It is not: it works by **photo-oxidative backbone scission, explicitly without recruiting host proteolytic machinery**. It must be excluded from any training or validation label set for this axis — it is a negative control, not an example.

What is still missing

- Flynn 2003 has never actually been read.** It defines the five ClpX recognition classes and holds the original trapped-substrate census. Its tables are main-text, behind the Cell Press paywall — an authenticated session should get them. Highest-value single fetch on the list.
- No *Klebsiella* protein has ever been chemically degraded.** The nearest result in any Gram-negative is ~40% depletion of GroEL in *E. coli*, by a plasmid-expressed peptide. This axis is a mechanistic prior, not a validated readout, and should be described that way in any paper.
- Around 40% of *E. coli* cytoplasmic proteolysis is attributable to none of ClpP, Lon or HslV.** So protease attribution is informative but incomplete — an unstable protein does not imply a handle we know about.
- No proteome-scale degron-tagging library exists for *E. coli*** — rechecked through 2026, including whether any CRISPRi library had been repurposed. The yeast equivalent covers >5,600 ORFs and the human one exists too, but both depend on ubiquitin machinery bacteria lack. So there is no direct ground truth for “tag this protein and it disappears”, only the indirect kind in Gupta's knockout columns.
- Almost nothing is curated.** MEROPS records **zero physiological ClpP substrates** for *E. coli* — the single entry is bovine insulin — plus 2 for Lon, 1 for FtsH and none at all for HslV. TopFIND's *E. coli* layer is UniProt electronic annotation with no literature behind it. There is no bacterial degron database of any kind. The absence is the opportunity, but it also means everything here has to be assembled from primary papers.
- Delivery and efflux are unsolved for this modality.** The cautionary number: purified *E. coli* ClpP is exquisitely sensitive to the ADEP class, yet whole-cell activity is >64 µg/mL, because ADEP is an efflux substrate — and the same pump is the dominant one in *Klebsiella*. DegP's compartment is attractive partly because it sidesteps the inner membrane entirely.

11 · SOURCES

The papers this rests on

Won HI, Zinga S, Kandrор O, et al. Targeted protein degradation in mycobacteria uncovers antibacterial effects and potentiates antibiotic efficacy. *Nat Commun* 15:4065 (2024). 10.1038/s41467-024-48506-8 — the 72-protein screen and the N-terminal-disorder result.

Izert-Nowakowska MA, Klimecka MM, Antosiewicz A, et al. Targeted protein degradation in *Escherichia coli* using CLIPPERS. *EMBO Rep* 26:3994–4016 (2025). 10.1038/s44319-025-00510-9 — the cytoplasmic handle, and the GroEL lesson.

Gupta M, Johnson ANT, Cruz ER, et al. Global protein turnover quantification in *Escherichia coli* reveals cytoplasmic recycling under nitrogen limitation. *Nat Commun* 15:5890 (2024). 10.1038/s41467-024-49920-8 — 3,262 proteins, protease attribution, and the disorder result at $p \approx 1e-208$.

Nagar N, Ecker N, Loewenthal G, et al. Harnessing machine learning to unravel protein degradation in *Escherichia coli*. *mSystems* 6:e01296-20 (2021). 10.1128/mSystems.01296-20 — the null result on motifs, and the network-connectivity finding.

Humbard MA, Surkov S, De Donatis GM, Jenkins LM, Maurizi MR. The N-degradome of *Escherichia coli*. *J Biol Chem* 288:28913–28924 (2013). 10.1074/jbc.M113.492108 — latent, proteolytically generated degrons.

Taylor EK, Barbosa PS, Kadambi T, et al. Molecular glue-like degraders of TEM β -lactamases by periplasmic protease DegP. *bioRxiv preprint* (2026). 10.64898/2026.03.30.715243 — the periplasmic handle.

Nie Q, Wu JW, Zhang K, et al. SsrA-based design of BacPROTACs for β -lactamase degradation in Gram-negative bacteria. *Chem Commun* 61:13149–13152 (2025). 10.1039/D5CC02015H — the compartment-mismatch cautionary tale.

Saunders RA, Stinson BM, Baker TA, Sauer RT. Multistep substrate binding and engagement by the AAA+ ClpXP protease. *PNAS* 117:28005–28013 (2020). 10.1073/pnas.2010804117 — the 5- and 20-residue thresholds. With **Kenniston JA, Baker TA, Sauer RT**, *PNAS* 102:1390–1395 (2005), 10.1073/pnas.0409634102 — the ~ 37 -residue reach.

Camberg JL, Viola MG, Rea L, et al. Location of dual sites in *E. coli* FtsZ important for degradation by ClpXP. *PLoS One* 9:e94964 (2014). 10.1371/journal.pone.0094964 — the internal linker degron.

Chai-Danino M, Ravensary-Modin N, Vladimirov VI, et al. Membrane-embedded polar residues target membrane proteins for degradation by the quality control protease FtsH. *Nat Commun* 17:3067 (2026). 10.1038/s41467-026-69829-8 — the inner-membrane handle.

Lin NM, Marino EC, Schlotmann JM, Rosen DA. Differential contributions of ClpX and ClpP to pulmonary virulence in classical and hypervirulent *Klebsiella pneumoniae*. *Infect Immun* 94(3) (2026). 10.1128/iai.00680-25 — the machinery is dispensable in Kp.

Van Damme P, et al. TRAINSPOTTER: profiling nascent protein N-termini indicative of bacterial translation initiation via deformylation-assisted N-terminomics. *Nucleic Acids Res* 54:gkag587 (2026). 10.1093/nar/gkag587 — 1,082 measured *E. coli* N-termini.

Jachmann C, et al. The *Escherichia coli* PeptideAtlas build. *J Proteome Res* 25:1027 (2026). 10.1021/acs.jproteome.5c00902 — 4,755 observed proteins and a proteome-wide peptide coordinate map.

Li GW, Burkhardt D, Gross C, Weissman JS. Quantifying absolute protein synthesis rates reveals principles underlying allocation of cellular resources. *Cell* 157:624 (2014). 10.1016/j.cell.2014.02.033 — the resynthesis-flux denominator.

Kshirsagar M, et al. Rapid and accurate prediction of protein homo-oligomer symmetry using Seq2Symm. *Nat Commun* 16:1969 (2025). 10.1038/s41467-025-57148-3 — assembly state at full coverage from sequence.

Iqbal N, Keller S, Ghanbarpour A. Structural plasticity of the membrane-bound protein degradation assembly. *Cell Rep* 45:117231 (2026). 10.1016/j.celrep.2026.117231 — FtsH is gated by an HflK/HflC cage.

Islam L, et al. ClpA- and ClpAP-catalyzed unfolding and translocation are differentially coupled to ATP binding. *Biophys J* (2026). 10.1016/j.bpj.2026.01.007 — for ClpAP, unfolding is rate-limiting.

González LJ, et al. Periplasmic protein quality control at atomic level in live cells. *Nat Commun* (2025). 10.1038/s41467-025-62340-6 — Prc cleaves a metallo- β -lactamase first, DegP then processes the fragments.

Han L, et al. Utilizing nanoinducers for precision degradation of bacterial protein. *Nat Commun* 16 (2025). 10.1038/s41467-025-66221-w — the SspB-adaptor route to ClpXP.

Morreale FE, Kleine S, Leodolter J, et al. BacPROTACs mediate targeted protein degradation in bacteria. *Cell* 185:2338–2353 (2022). 10.1016/j.cell.2022.05.009 — the original modality, and the one that does not transfer.